

```
# indicate the rack and dc for this node
dc=DC1
rack=RAC1
```

- iv. After completing the previous step, start the seed nodes (e.g., 110.82.155.0, 110.82.155.2) one by one, and then start the remaining nodes (e.g., 110.82.155.1).

```
$ sudo service cassandra start
```

To verify the installation, use the following command:

```
$ nodetool status
```

Optimisation of Apache Cassandra

Cassandra is well known for its good write performance. However, there are a few cases where things do work but could be better. Here are a few recommendations to optimise the performance of Apache Cassandra.

Tuning the Java resources: You can optimise the performance of Cassandra by tuning Java heap size. The default configuration of the heap size is based on the system memory, as shown below:

System memory	Heap size
Less than 2GB	1/2 of system memory
2GB to 4GB	1GB
Greater than 4GB	1/4 system memory, but not more than 8GB

If you are planning to change the heap size, it's recommended that you change `MAX_HEAP_SIZE` and `HEAP_NEWSIZE` together.

MAX_HEAP_SIZE: This sets the maximum heap size for the JVM. The same value is also used for the minimum heap size. This allows the heap to be locked in memory at the start of the process, to keep it from being swapped out by the OS.

HEAP_NEWSIZE: This is the size of the young generation heap. The larger this is, the longer Garbage Collection pause times will be. The shorter it is, the more expensive Garbage Collection will be (usually). A good guideline is 100MB per CPU core.

By using the cache efficiently: You should consider key

cache and row cache if rows do not belong to the large tables. The latest version of Cassandra allows us to configure partial or full caching of each partition (`rows_per_partition`).

Here are some tips to use the cache efficiently:

- Store lower-demand data or data with extremely long partitions in a table with minimal or no caching.
- Deploy a large number of Cassandra nodes under a relatively light load per node.
- Logically separate heavily-read data into discrete tables.

Improve write performance by configuring memtable throughput: Configuring *memtable* throughput can improve write performance. Cassandra flushes *memtables* to disk, creating SSTables when the commit log space threshold has been exceeded. Configure the commit log space threshold per node in `cassandra.yaml`. How you tune *memtable* thresholds depends on your data and write load.

Enable compression to optimise for read performance:

A proper compression algorithm helps to increase the performance of Cassandra. As per the official documentation, writes on compressed tables can show up to a 10 per cent performance improvement.

The compression algorithms are *LZ4Compressor*, *SnappyCompressor*, and *DeflateCompressor*.

You must also choose the appropriate compaction strategy for the corresponding I/O pattern. 

References

- [1] <http://docs.datastax.com/en/cassandra/3.x/cassandra/install/installTOC.html>
- [2] <http://www.planetcassandra.org/cassandra/>
- [3] http://docs.datastax.com/en/cassandra_wm/3.x/cassandra/install/InstallWin.html
- [4] <http://www.planetcassandra.org/Try-Cassandra/>
- [5] <http://docs.datastax.com/en/cassandra/3.x/cassandra/initialize/initSingleDS.html>
- [6] <http://docs.datastax.com/en/cassandra/3.x/cassandra/operations/opsMonitoring.html>

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